

Initial Project Plan (week 10, submission date: 5 June 2026)

Group Name	Group 11																				
Members	<table><tr><td>Full Name</td><td>Matric Number</td><td>Email</td></tr><tr><td>LOGITH A/L MOHANA KRISHNAN</td><td>223021</td><td>223021@student.upm.edu.my</td></tr><tr><td>SASHVHANT A/L VADEVELL</td><td>223726</td><td>223726@student.upm.edu.my</td></tr><tr><td>HASSAN SAED</td><td>213870</td><td>213870@student.upm.edu.my</td></tr><tr><td>XU CHENHUI</td><td>217944</td><td>217944@student.upm.edu.my</td></tr><tr><td>QIAN YIJIA</td><td>225842</td><td>225842@student.upm.edu.my</td></tr></table>			Full Name	Matric Number	Email	LOGITH A/L MOHANA KRISHNAN	223021	223021@student.upm.edu.my	SASHVHANT A/L VADEVELL	223726	223726@student.upm.edu.my	HASSAN SAED	213870	213870@student.upm.edu.my	XU CHENHUI	217944	217944@student.upm.edu.my	QIAN YIJIA	225842	225842@student.upm.edu.my
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Problem scenario description	A severe typhoon has completely cut off a remote island community in Langkawi, destroying the local seaport and making maritime access impossible. The only way to deliver critical emergency supplies is via helicopter airdrop. The rescue helicopter has a strict maximum weight capacity of 1000 kg. Responders have a stockpile of various supply crates, each with a specific weight and an assigned survival priority value.																				
Why it is important	Maximizing the life-saving impact of the limited payload is essential. Guessing or packing suboptimally wastes crucial helicopter space, leading to a direct loss in survival priority points which represents a failure to deliver critical medicine, food, or communication gear to stranded victims.																				
Problem specification	The disaster relief payload is modeled as a 0/1 Knapsack Problem. There are 6 indivisible supply crates, each with a weight (w_i) and priority value (v_i). The absolute constraint is the 1000 kg helicopter capacity limit (W). The mathematical goal is to maximize the objective function $\sum v_i x_i$ subject to $\sum w_i x_i \leq$, where $x_i \in$.																				

Potential solutions	We evaluated Brute Force, Greedy Algorithms, and Dynamic Programming. We selected the Dynamic Programming paradigm because human intuition or a Greedy approach (sorting by value-to-weight ratio) fails and traps the algorithm due to the 0/1 constraint. DP guarantees the absolute maximum survival value by evaluating overlapping sub-capacities.
Sketch (framework, flow, interface)	

Project Proposal Refinement (week 11, submission date: 12 June 2026)

Group Name	Group 11
Problem statement	Determine the optimal combination of indivisible supply crates to load onto a rescue helicopter to maximize total survival priority value without exceeding the 1000 kg structural weight limit.
Objectives	1. To accurately model the physical disaster relief payload constraint mathematically using binary variables (x_i). 2. To apply Dynamic Programming to populate a 2D array resolving overlapping sub-capacities to find the global optimum. 3. To demonstrate why a standard Greedy Algorithm fails by falling into a local optimum trap, validating the DP choice.

Expected output	1. The calculated absolute maximum total survival priority value (1300). 2. A detailed sequence of the specific crates selected utilizing exactly 1000 kg. 3. Execution time metrics comparing efficiency.																												
Problem scenario description	A severe typhoon has completely cut off a remote island community in Langkawi, destroying the local seaport and making maritime access impossible. The only way to deliver critical emergency supplies is via helicopter airdrop. The rescue helicopter has a strict maximum weight capacity of 1000 kg.																												
Why it is important	Maximizing the life-saving impact of the limited payload is essential. Guessing or packing suboptimally wastes crucial helicopter space, leading to a direct loss in survival priority points.																												
Problem specification	Modeled as a 0/1 Knapsack Problem. Absolute constraint: 1000 kg capacity limit (W). Mathematical goal: maximize $\sum v_i x_i$ subject to $\sum w_i x_i \leq$.																												
Potential solutions	Evaluated Brute Force, Greedy, and Dynamic Programming. Selected Dynamic Programming to guarantee absolute maximum survival value.																												
Sketch (framework, flow, interface)	<table><tr><th>Item Index (i)</th><th>Crate Content</th><th>Weight (wi) in kg</th><th>Priority Value (vi)</th></tr><tr><td>0</td><td>Heavy Surgical Equipment</td><td>600</td><td>800</td></tr><tr><td>1</td><td>High-Yield Water Purifiers</td><td>500</td><td>650</td></tr><tr><td>2</td><td>Concentrated Emergency Rations</td><td>500</td><td>650</td></tr><tr><td>3</td><td>Portable Diesel Generators</td><td>400</td><td>500</td></tr><tr><td>4</td><td>First Aid Trauma Kits</td><td>100</td><td>150</td></tr><tr><td>5</td><td>Satellite Communication Gear</td><td>100</td><td>120</td></tr></table>	Item Index (i)	Crate Content	Weight (wi) in kg	Priority Value (vi)	0	Heavy Surgical Equipment	600	800	1	High-Yield Water Purifiers	500	650	2	Concentrated Emergency Rations	500	650	3	Portable Diesel Generators	400	500	4	First Aid Trauma Kits	100	150	5	Satellite Communication Gear	100	120
Item Index (i)	Crate Content	Weight (wi) in kg	Priority Value (vi)																										
0	Heavy Surgical Equipment	600	800																										
1	High-Yield Water Purifiers	500	650																										
2	Concentrated Emergency Rations	500	650																										
3	Portable Diesel Generators	400	500																										
4	First Aid Trauma Kits	100	150																										
5	Satellite Communication Gear	100	120																										

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PS C:\Users\hassan saed> java AerialReliefOptimization
=== Aerial Relief Airdrop Optimization ===
Maximum Survival Priority Value Achieved: 1300
Total Payload Weight Used: 1000 kg / 1000 kg

Selected Supply Crates to Load:
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Item Index 1 | High-Yield Water Purifiers      | Weight: 500 kg | Value: 650
Item Index 2 | Concentrated Emergency Rations      | Weight: 500 kg | Value: 650
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Execution Time (Core Algorithm): 967599 ns (0.9676 ms)
PS C:\Users\hassan saed>

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Methodology

Milestone	Time
Scenario formulation, problem definition (Langkawi typhoon, 1000kg helicopter limit), and initial UI/UX sketching.	Wk 10
Algorithm paradigm comparison, selection of Dynamic Programming (0/1 Knapsack), and defining methodology.	Wk 11
Java programming: implementing 2D array data structures and writing the core Dynamic Programming logic. Debugging.	Wk 12
Conduct analysis of correctness, evaluate time complexity $O(n \cdot W)$, and execute comparative Greedy algorithm testing.	Wk 13
Prepare online portfolio, compile documentation, and structure final presentation deck.	Wk 14

Project Progress (Week 10 - Week 14)

Milestone 1	Planning and Design Phase
Date (week)	Week 10 - Week 11
Description/ sketch	Finalized the remote island typhoon scenario, defined the Knapsack variables (6 crates, 1000kg limit), completed algorithm comparisons, and submitted the initial project plan and refined proposals.
Role	LOGITH (Defined constraints, defined HMW questions, managed submissions)
	SASHVHANT (Compared paradigms, justified DP, drafted pseudocode)
	HASSAN (Researched Java DP implementations)
	XU (Created conceptual map, identified DP recurrence)
	QIAN (Defined edge-cases, planned portfolio layout)

Milestone 2	Implementation and Finalization Phase
Date (Wk)	Week 12 - Week 14
Description/ sketch	Translated pseudocode into Java, analyzed time complexity, integrated content into the online portfolio, and prepared the final presentation deck.
Role	LOGITH (Monitored progress, ensured equitable workload and finalized documentation)
	YIJIA (Verified output against manual traces, documented methodology)
	SASHVHANT (Wrote, debugged, and compiled the final Java program)
	XU (Executed asymptotic time and space complexity analysis)
	HASSAN (Built online portfolio, integrated code/diagrams)